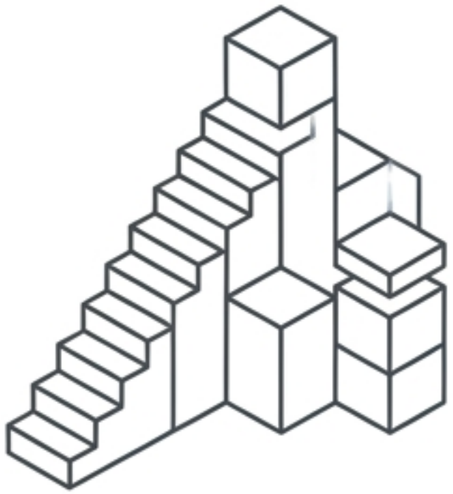
Blue Ocean Waves

Making the impossible a reality with advanced SAR integration and tactical edge delivery.

Gemini Notebook

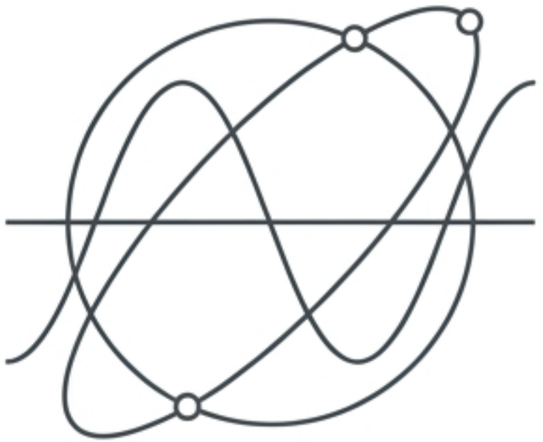
The AI Says No

Mainstream AI models require pristine, pre-processed optical data. When faced with raw, noisy wave data, conventional systems fail. They say it can’t be done.



The Incremental Alchemy

We solve the chicken before the egg. If we cannot find the entire path at once, we engineer baby steps. Atomic, incremental advancements that stack into impossible realities.



The Medium

We concentrate on waves. By illuminating existing, daily satellite data with complex combinations, we unlock a massive blue ocean of operational intelligence.

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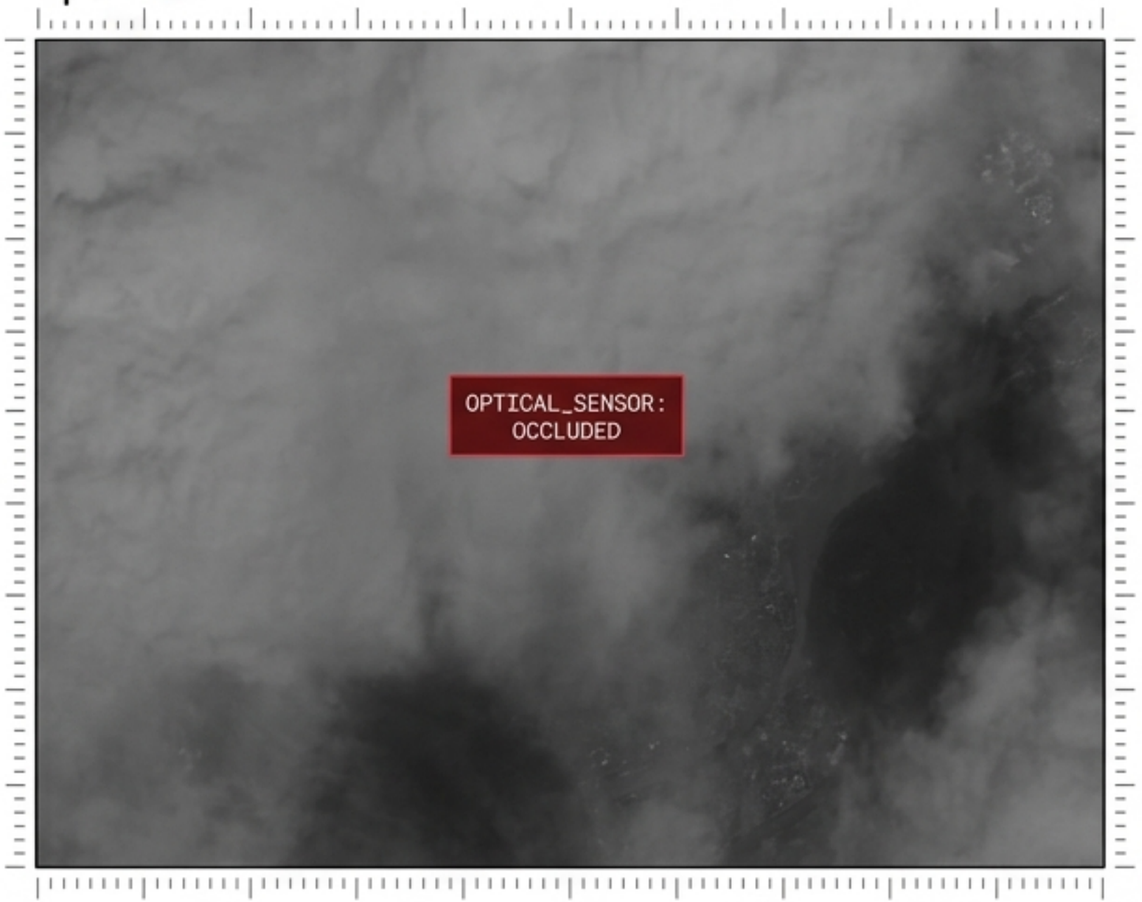
Illuminating the unseen through the physics of waves.

Optical



Optical sensors capture light. They are defeated by clouds, weather, and night.

SAR Waves

SAR sensors capture physical structure. They pierce through the occlusion to map the fundamental geometry of the Earth, 24/7.

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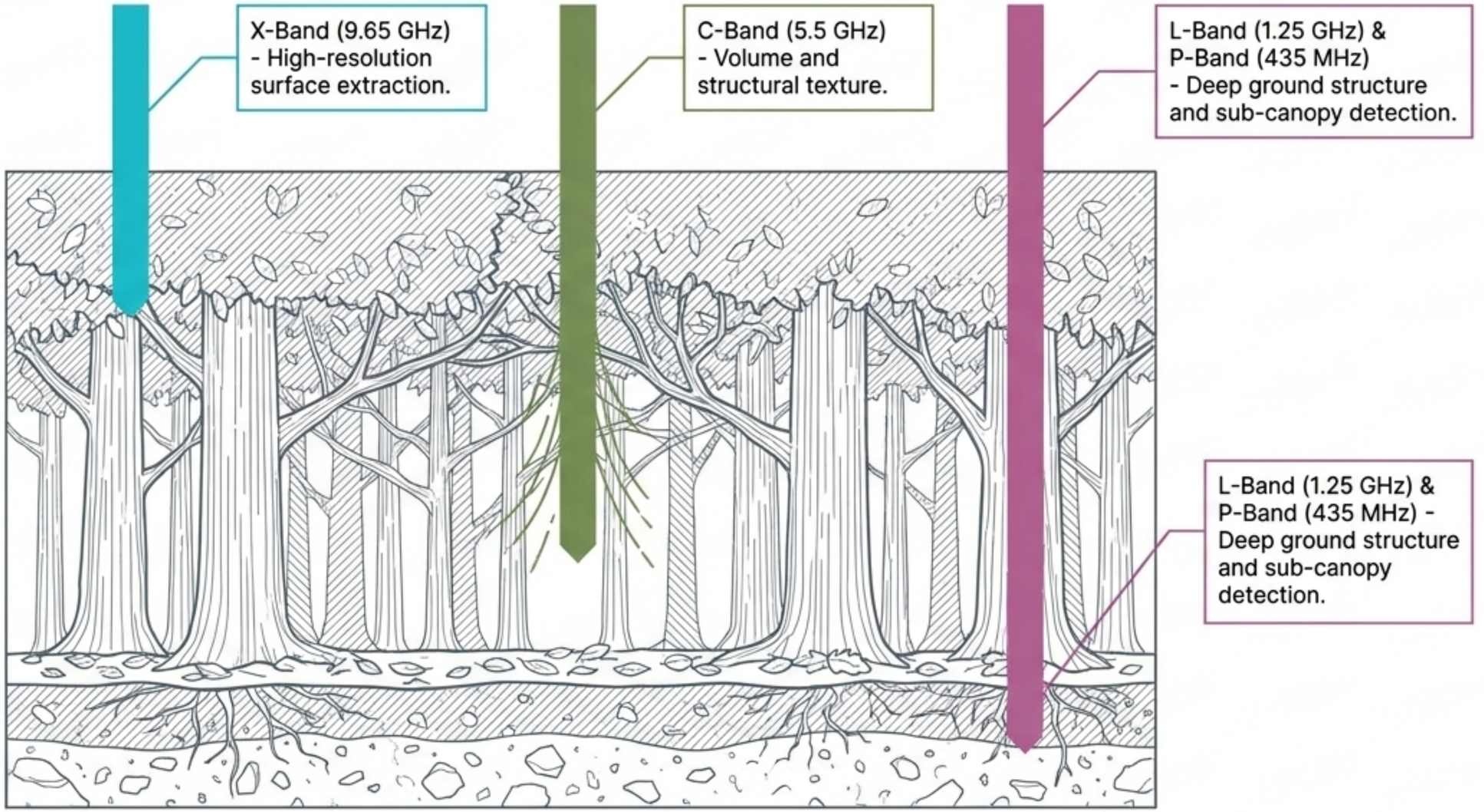
Stacking frequencies to peel back the Earth's layers.



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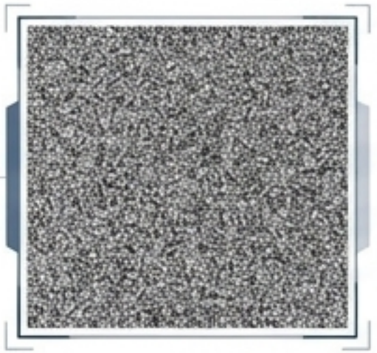
The SAR Triad of Enemies.

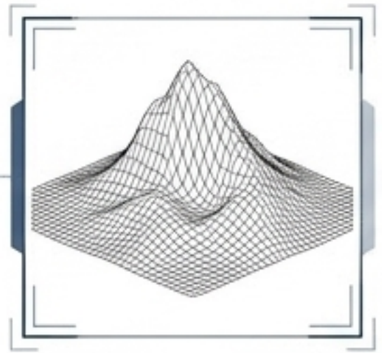


GATE 1

Speckle Noise

STATUS: CRITICAL INTERFERENCE

Coherent interference causing granular distortion.
Conventional AI blurs this into uselessness.

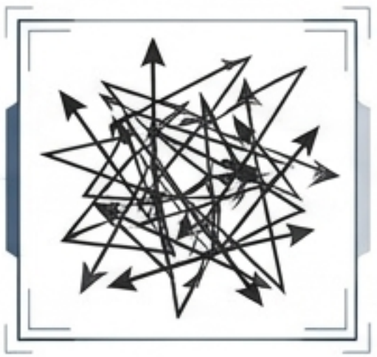


GATE 2

Geometric Distortion

LAYOVER / SHADOW

Side-looking radar geometry warps steep terrain
into layover and shadow blind spots.



GATE 3

Backscatter Chaos

SIGNAL RETURNS: COMPLEX / NON-LINEAR

Complex signal returns that optical-trained foundation
models cannot decipher.

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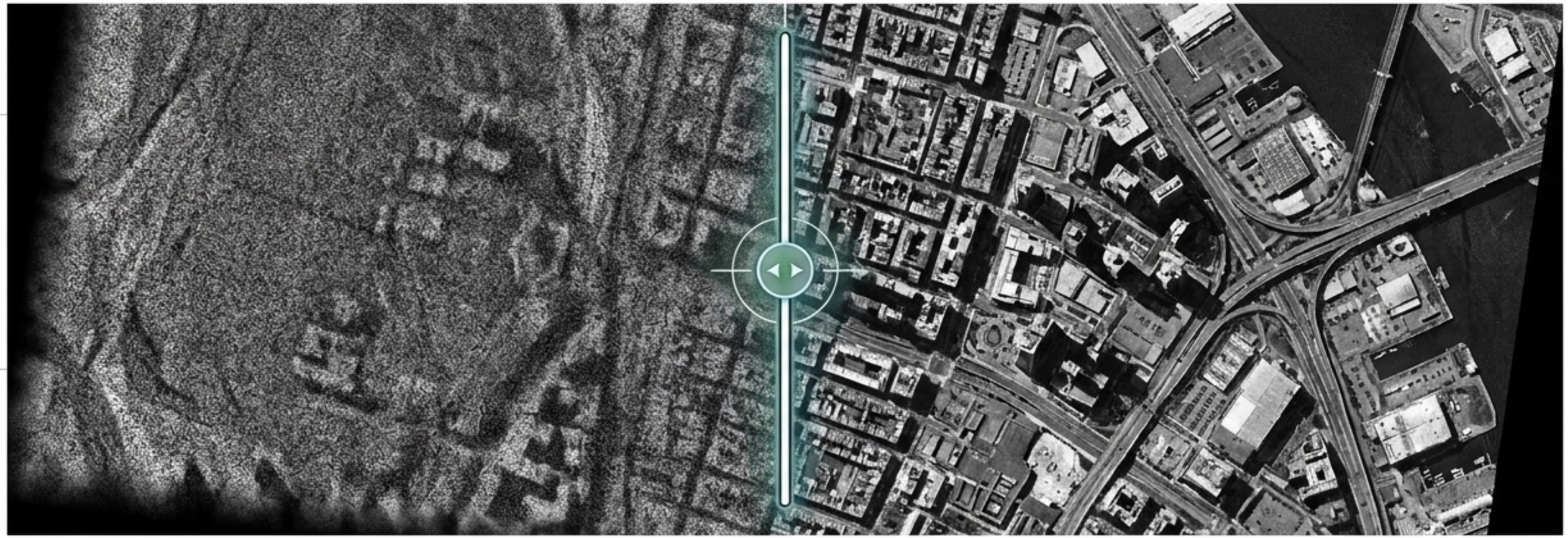
Baby Step 1: Denoising the void without losing the signal.



EXECUTE: Denoising Diffusion Probabilistic Models (DDPM) - Reconstructing the signal iteratively from Gaussian noise.
EXECUTE: DenoDet V2 (Phase-Amplitude Cross Denoising) - Using the inherent stability of phase information to guide the feature denoising of amplitude.
STATUS: We do not blur data. We reconstruct the physical reality beneath the noise.

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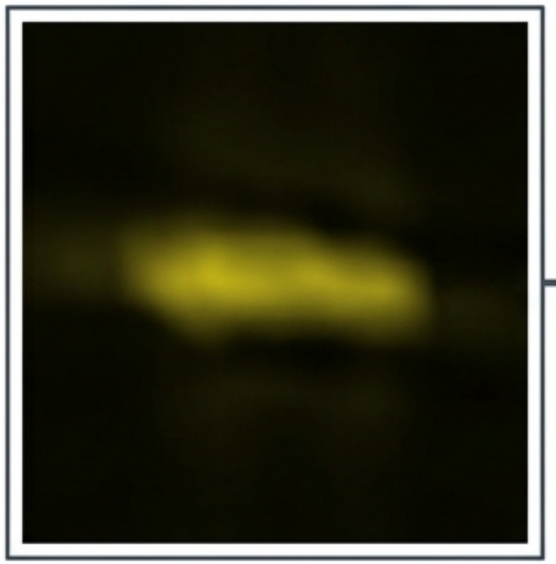
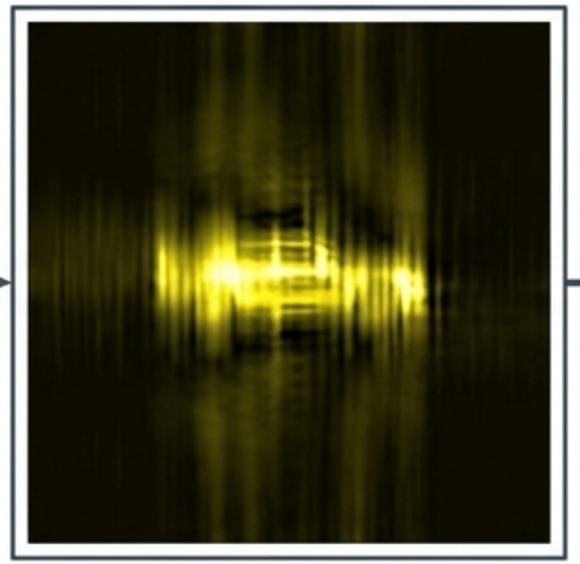
Baby Step 2: Forcing focus from scattered history.

Raw Phase History



Partially Focused



Perfectly Focused

Raw Phase History → **Time-Domain Back-Projection (TDBP)** → **Quantum Range-Doppler** → **Focused Target Extraction**

Leveraging advanced sparsity-driven reconstruction and Time-Domain Back-Projection (TDBP) to achieve **sub-meter spotlight focusing**, compensating for **motion errors** in the dark.

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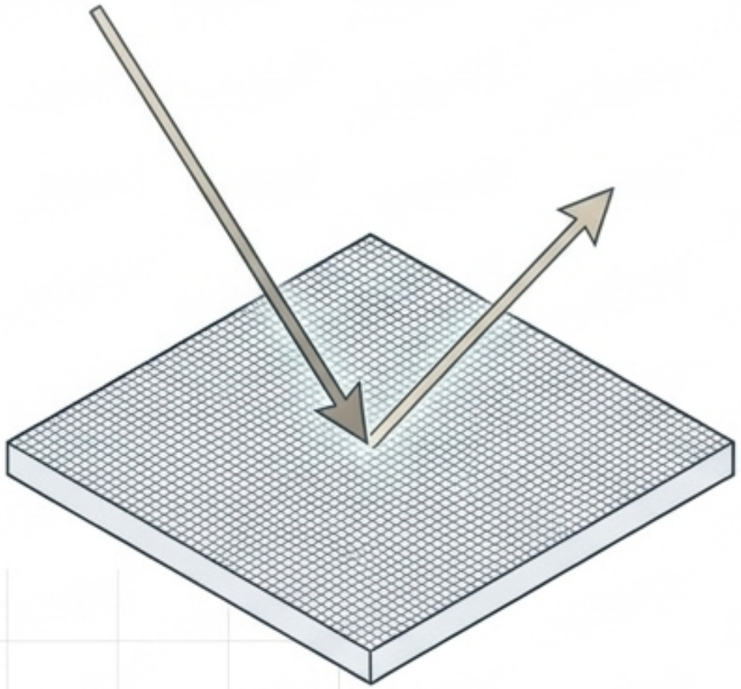
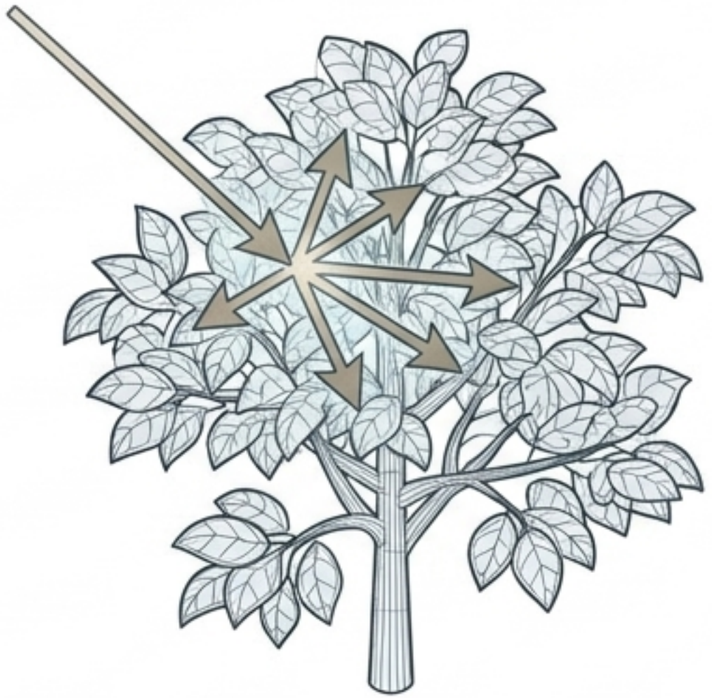
Decoding the Backscatter.



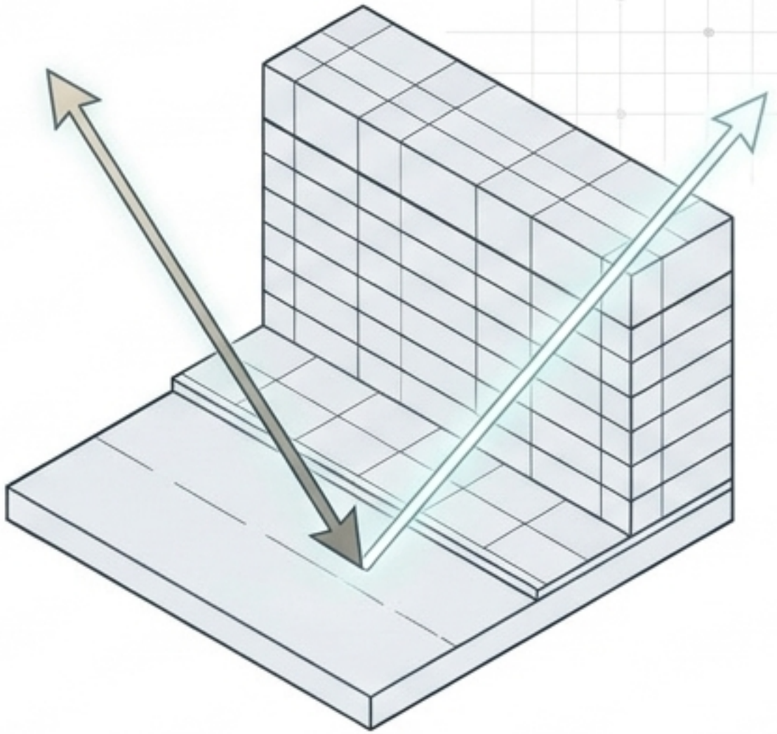
Direct Backscatter

Represents dark, smooth surfaces like runways or water. Signals bounce away, leaving the area black.



Volume Scattering

Represents vegetation density. Radar waves bounce internally within complex volumes, returning a textured mid-tone.



Double-Bounce

Represents man-made geometry. The right angle acts as a corner reflector, returning an violently bright, undeniable signal.

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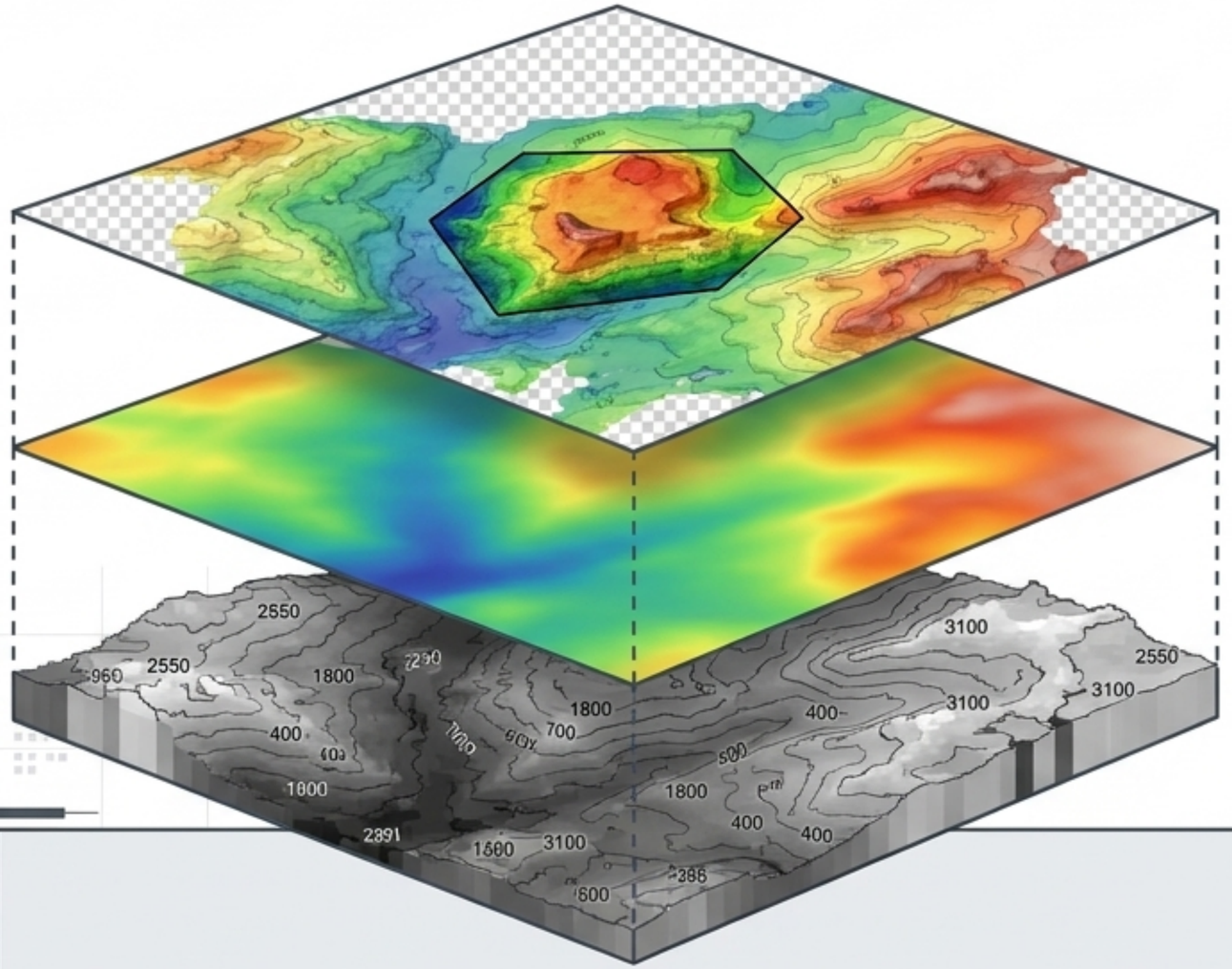
Raster Symbology: Coloring the invisible.



- **PROCESS: Singleband Pseudocolor** - Applying aggressive contrast enhancements to stretch values to absolute Min/Max limits.
- **PROCESS: Custom Transparency Options** - Stripping away 100% of non-essential corner pixels to isolate critical target geometry.
- **OUTPUT**: Data that speaks to the human eye, ready for tactical overlay.

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The Orbital Arsenal: Multi-Constellation Architecture.

Constellation	Band	Frequency	Capability
Sentinel-1 (ESA)	C-Band	5.5 GHz	Wide-swath surface roughness.
NISAR (NASA-ISRO)	L/S-Band	1.25/3.2 GHz	Deep canopy SweepSAR penetration.
TerraSAR-X / TanDEM-X	X-Band	9.65 GHz	Formation-flying interferometry.
Capella Space & Umbra	X-Band	Up to 1200 MHz	Sub-meter Spotlight vehicle detection.
BIOMASS	P-Band	435 MHz	Absolute forest floor baseline mapping.

We do not rely on a single sensor. We fuse the world's public and commercial constellations to achieve persistent, multi-frequency illumination.

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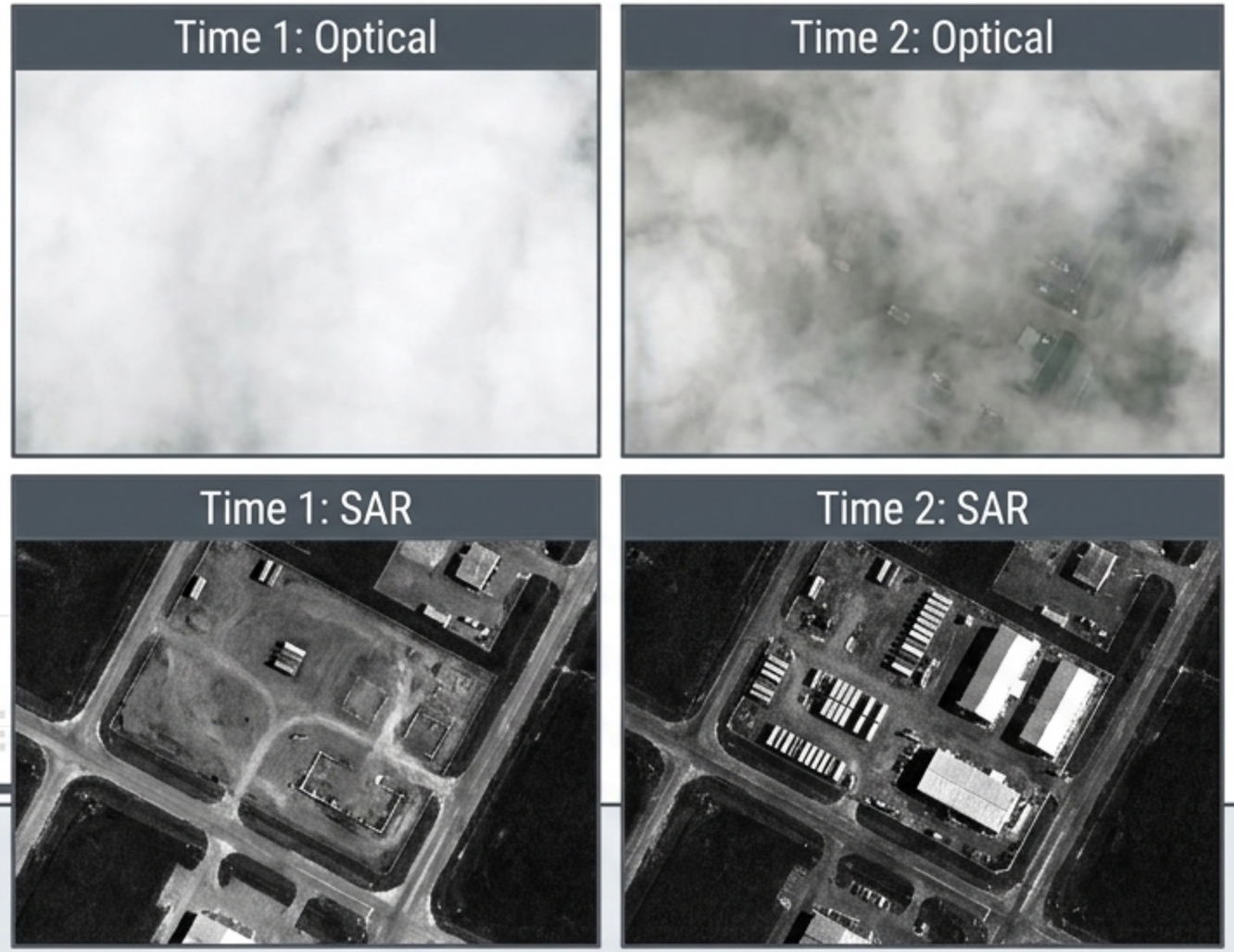
SkyCap: Bitemporal optical-SAR fusion.



- When optical fails, **Amplitude Change Detection (ACD)** prevails.
- By co-registering optical SkySat data with Capella Space SAR imagery, we track sub-meter infrastructure changes changes in any weather, completely bypassing the optical bottleneck.

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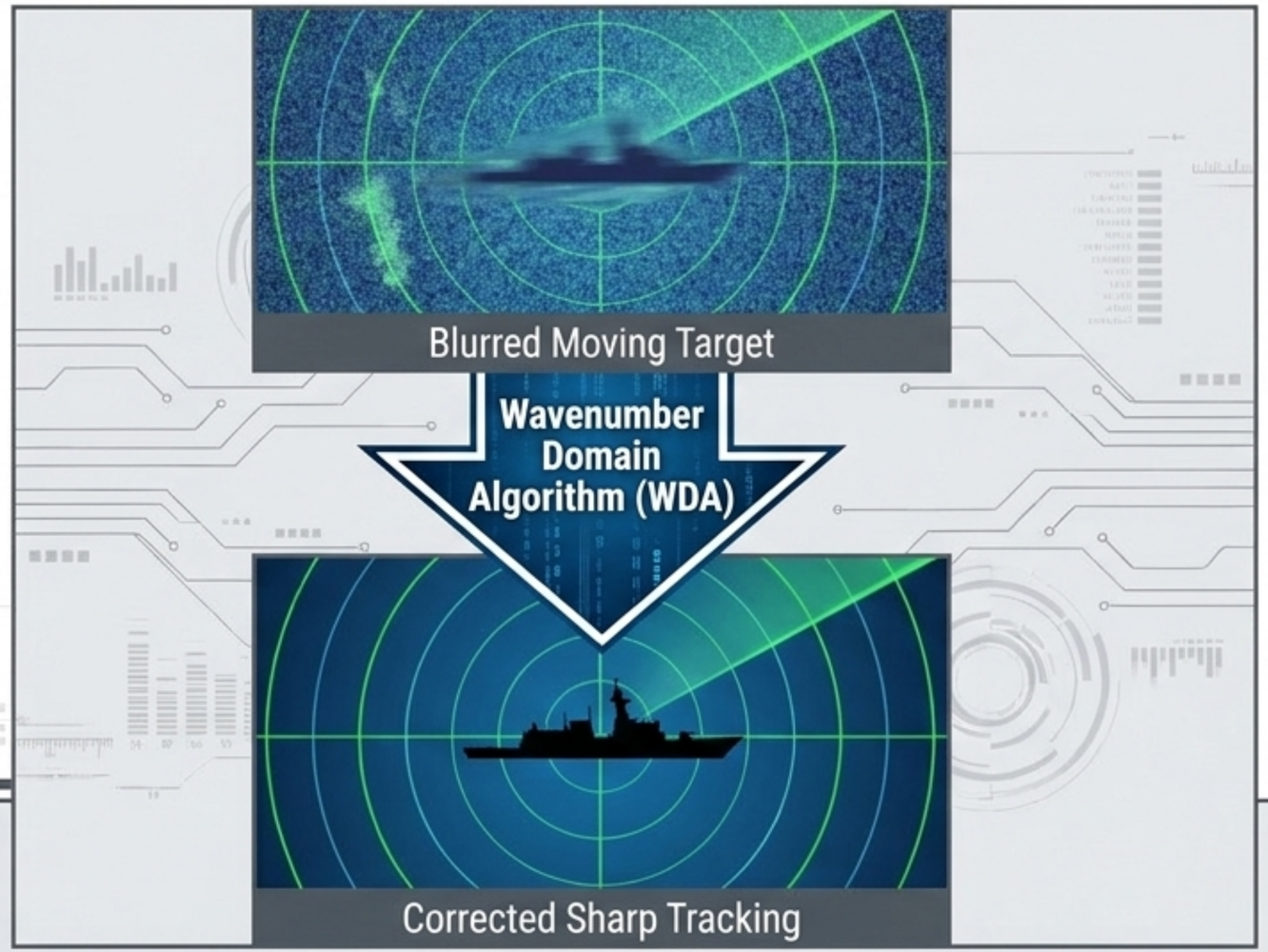
Pushing the limits: Moving target imaging.



- Standard SAR processing assumes a static Earth. Moving targets blur and displace.
- **EXECUTE: Wavenumber Domain Algorithm (WDA)**
- By applying deep Time-Domain Back-Projection and FMCW phase compensation, we reverse range cell migration. The result? We track moving vessels, vehicles, and assets from orbit.

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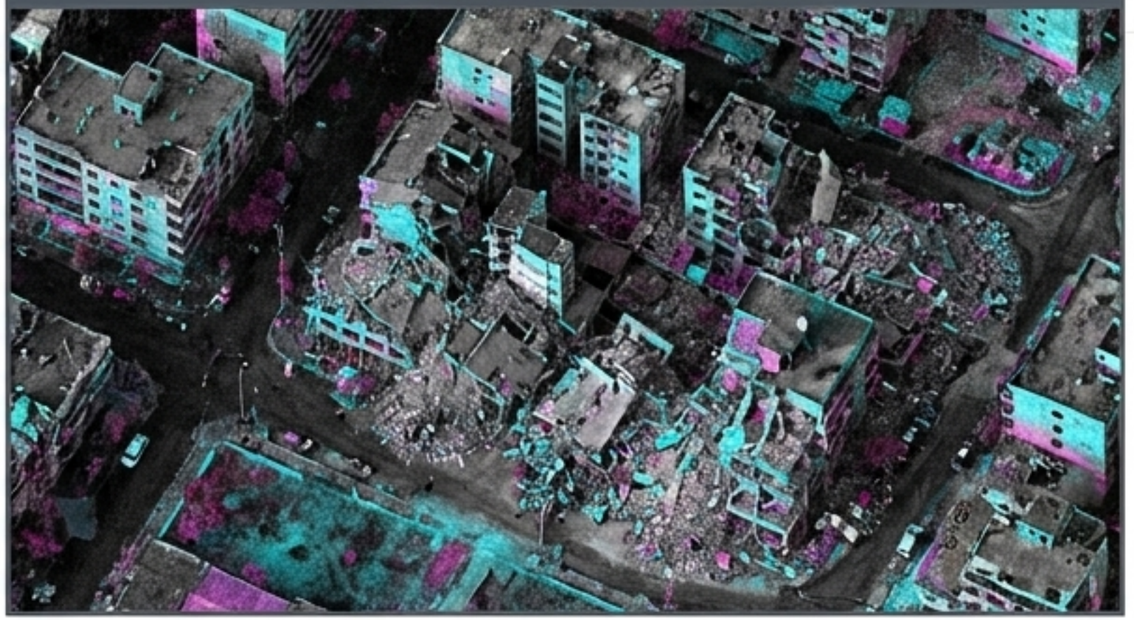
The Impossible Challenge: Post-event single image detection.

MISSING:
PRE-EVENT DATA

POST-EVENT SINGLE IMAGE



- **The Orthodoxy:** Disaster damage mapping requires a ‘Before’ and ‘After’ image to detect change.
- **The Reality:** In blue ocean scenarios, pre-event VHR SAR data simply doesn’t exist.
- **The Mission:** Extract collapsed buildings from a single, noisy, post-earthquake SAR image. The AI says it cannot be done.

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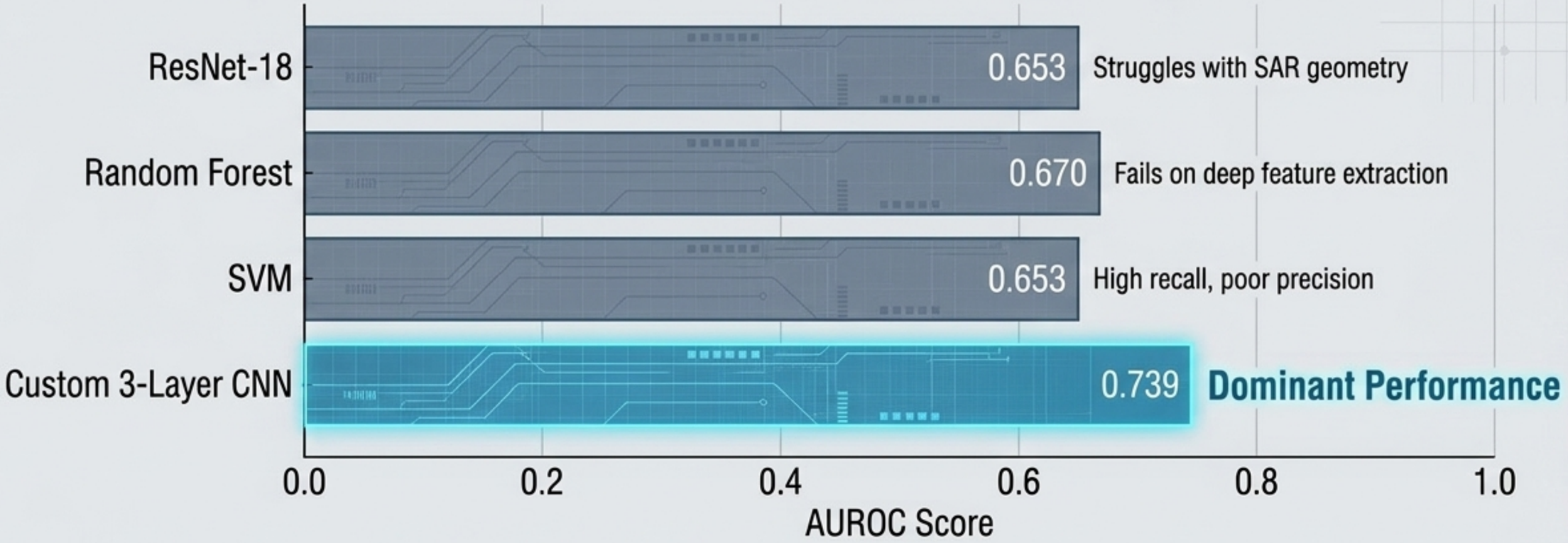
Algorithm performance in the dark.



Complex, heavyweight optical networks (ResNet) fail on SAR. By utilizing a tailored CNN, we efficiently capture the unique structural patterns of raw SAR backscatter, successfully mapping destroyed infrastructure without a baseline.

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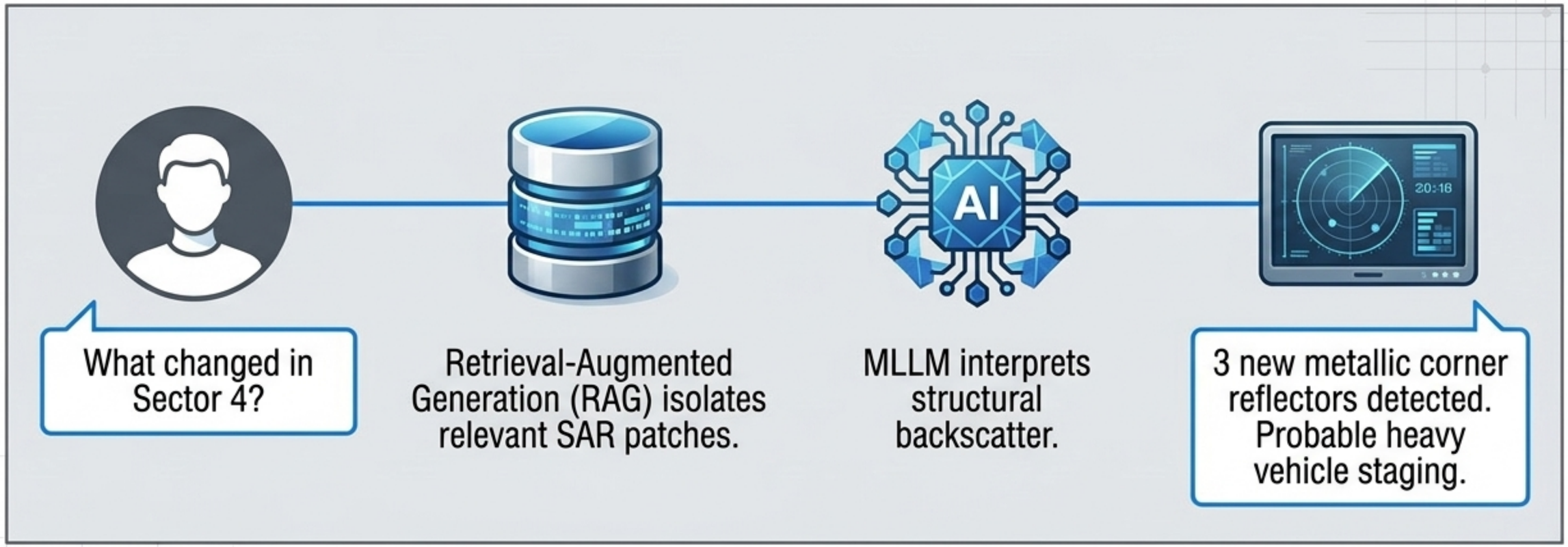
SAR-RAG: Conversing with the invisible.



We bridge the gap between complex orbital physics and tactical decision-makers. SAR-RAG integrates external spatial knowledge directly into the inference engine, allowing operators to query satellite imagery in plain English.

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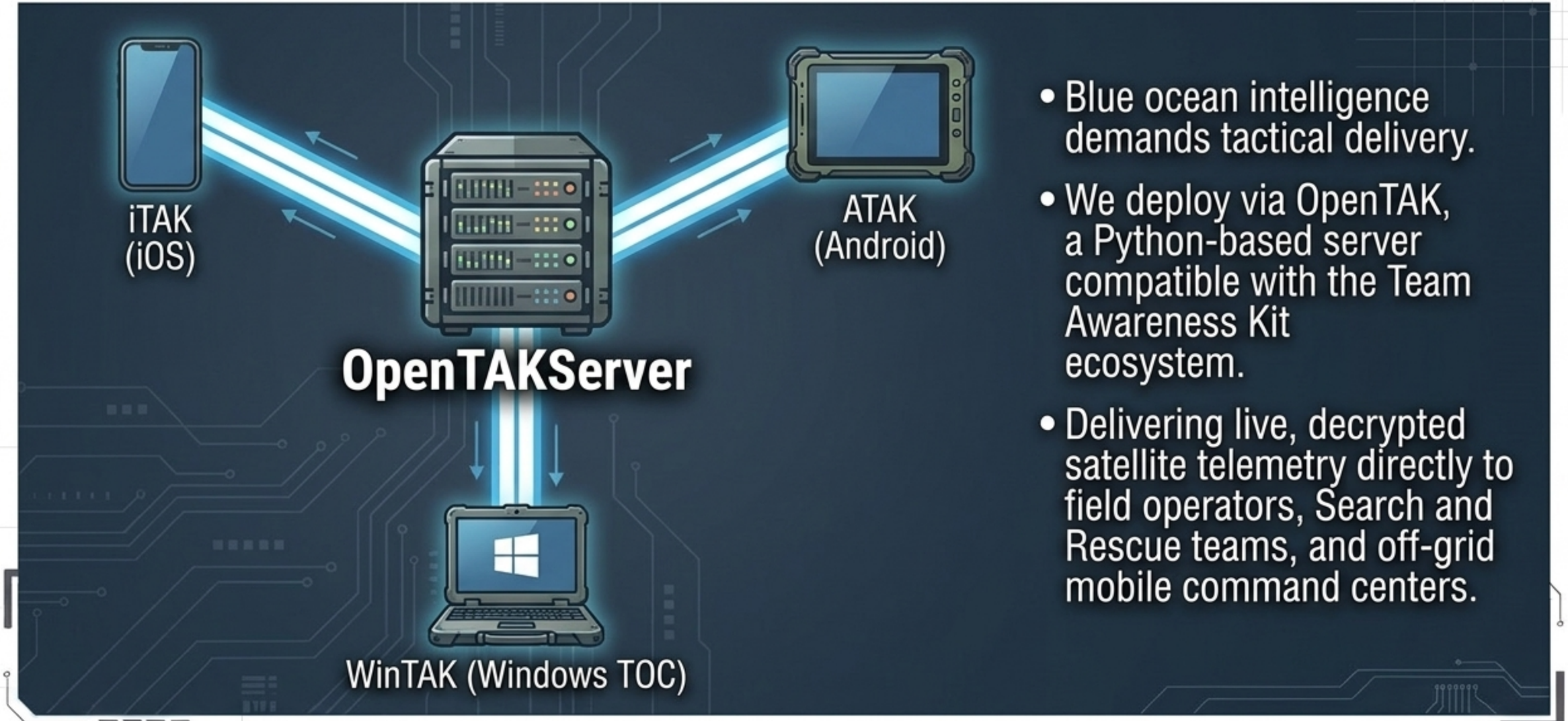
The ‘Deep Web’ Tactical Edge: OpenTAKServer.



- Blue ocean intelligence demands tactical delivery.
- We deploy via OpenTAK, a Python-based server compatible with the Team Awareness Kit ecosystem.
- Delivering live, decrypted satellite telemetry directly to field operators, Search and Rescue teams, and off-grid mobile command centers.

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Cursor-on-Target (CoT): The DNA of tactical messaging.

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   stale="2024-05-15T10:35:00Z"
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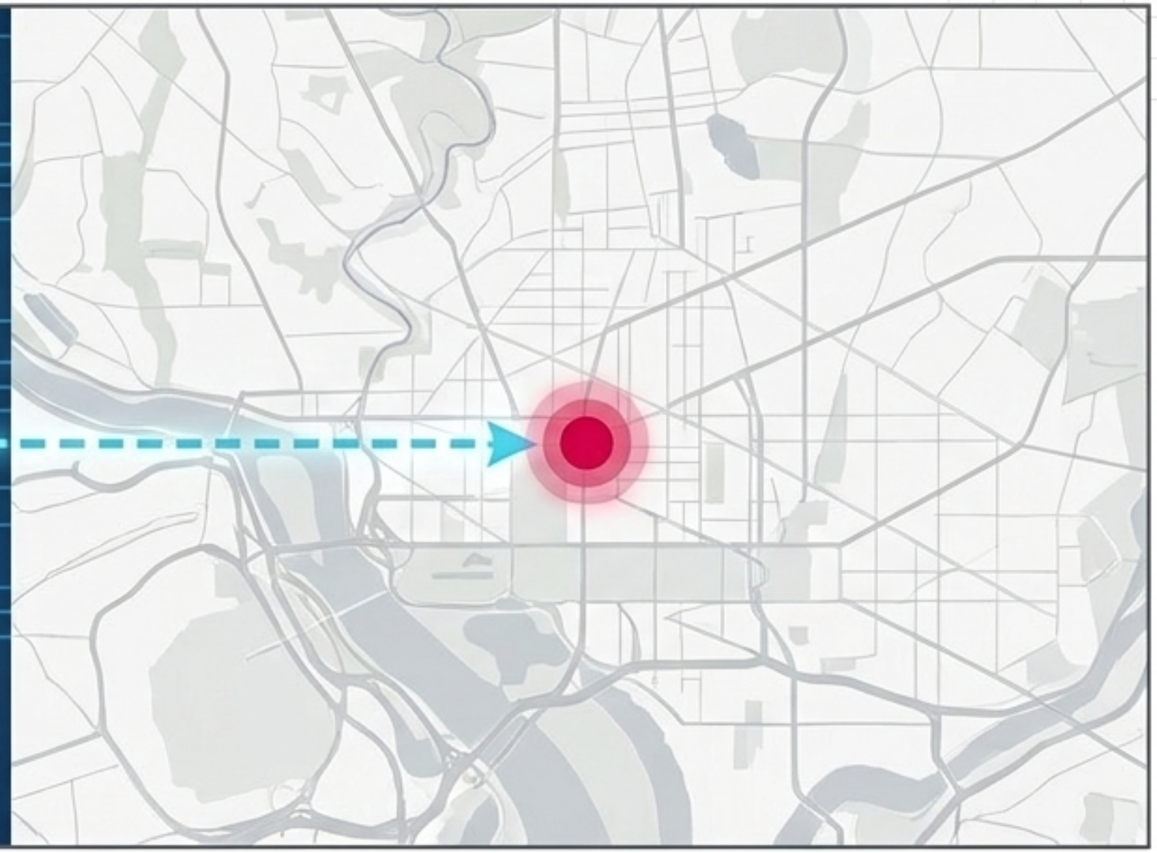
   </detail>
</event>
```

WHAT: The type of object (SAR anomaly, vehicle, hazard).

WHERE: Precise geospatial coordinates extracted from SAR phase history.

WHEN: Timestamp of valid data transmission.



We translate highly complex, multi-gigabyte orbital radar signals into a single, actionable XML packet that instantly appears on an operator's mobile device.

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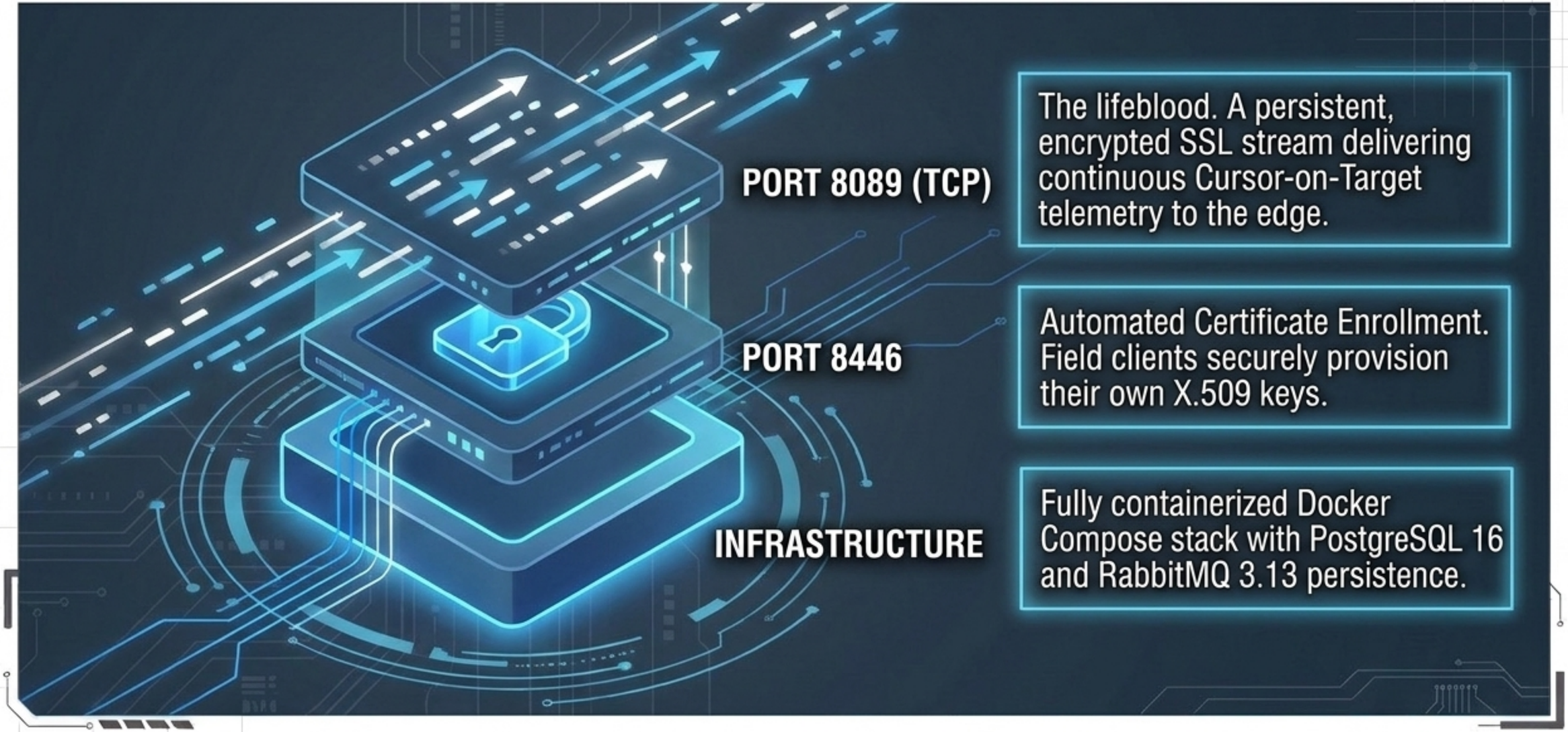
Security & Reliability in the Dark.



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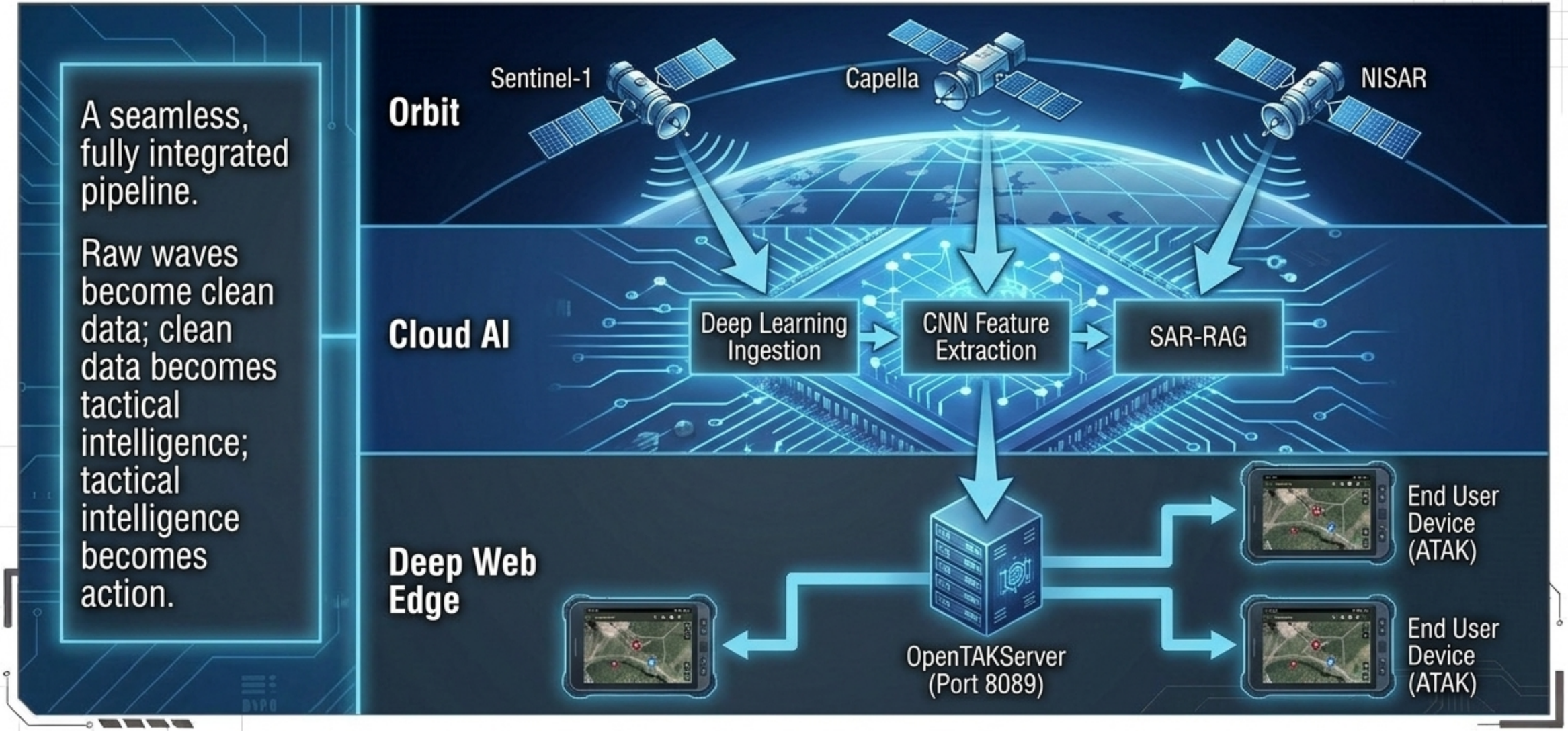
System Topology: From orbit to the absolute edge.



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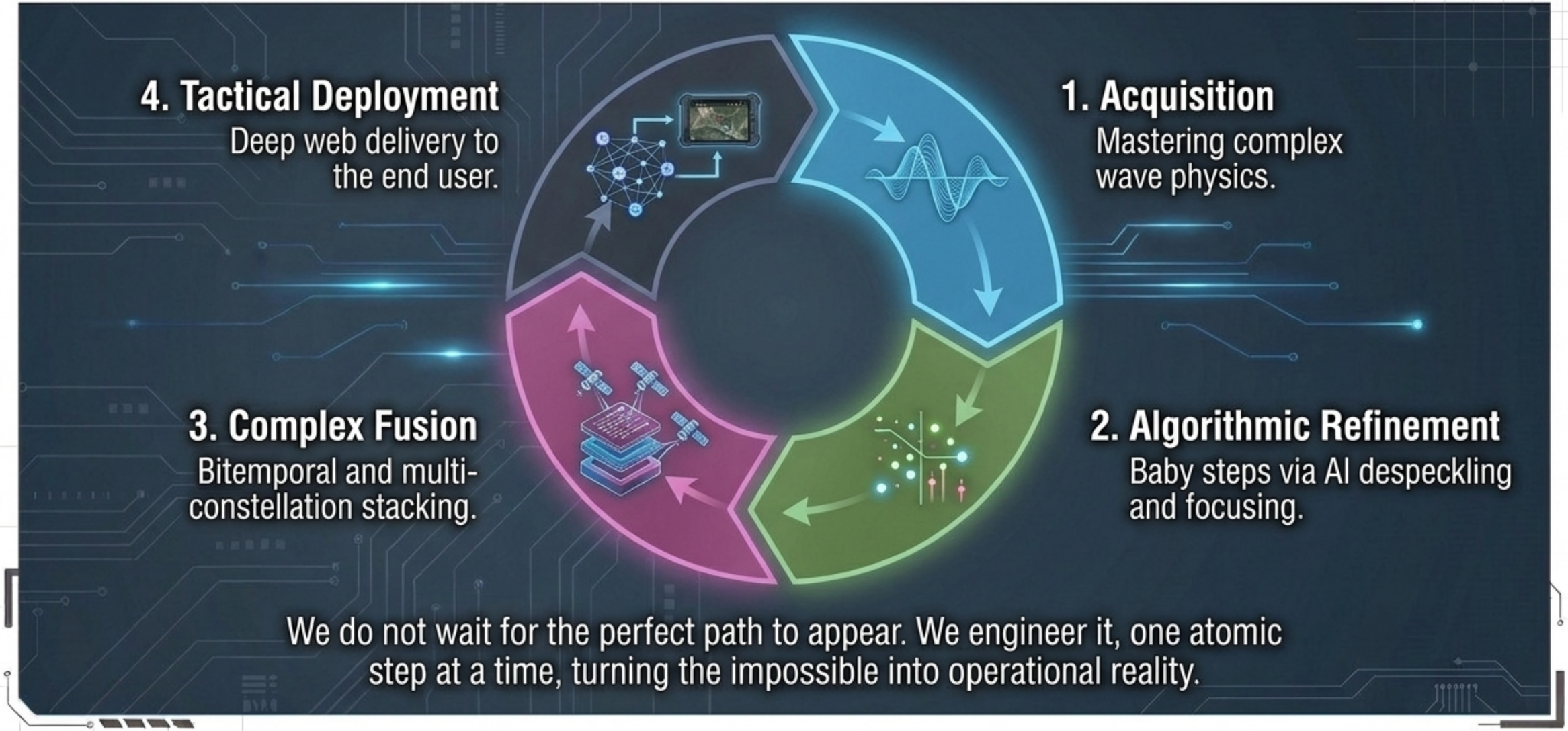
The proof is in the increments.



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// END_TRANSMISSION
// ALL_SYSTEMS_NOMINAL

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